

EmpathyGym

Voice-first customer support training simulator
powered by AI

<https://empathy-gym-c5f7a.web.app>

100% Free to Host

Gemini 2.5 Flash AI

Firebase Firestore

React + TypeScript

Browser-Native Voice

Real-time Scoring

What Is EmpathyGym?

EmpathyGym is a voice-first customer support training simulator that runs entirely in the browser. Trainees speak to an AI-powered customer persona in real time, receive live feedback on the customer's frustration level, and get a scored performance report at the end of every session — with zero backend infrastructure and zero monthly cost.

Use Cases

Five high-impact scenarios where EmpathyGym delivers measurable value.

New Agent Onboarding

Replace expensive human role-play with unlimited AI-powered practice sessions. New agents complete 10 persona scenarios before their first live call, cutting onboarding time from days to hours.

Empathy Skill Remediation

Managers assign specific personas that mirror an agent's weak spots. The agent practices until their Empathy Score consistently exceeds 75%, with objective AI-generated coaching tips each session.

Product Launch Preparation

Before a major launch, support leads create personas matching anticipated complaint types and run the whole team through targeted sessions — reducing first-call escalations significantly.

Self-Directed Continuous Learning

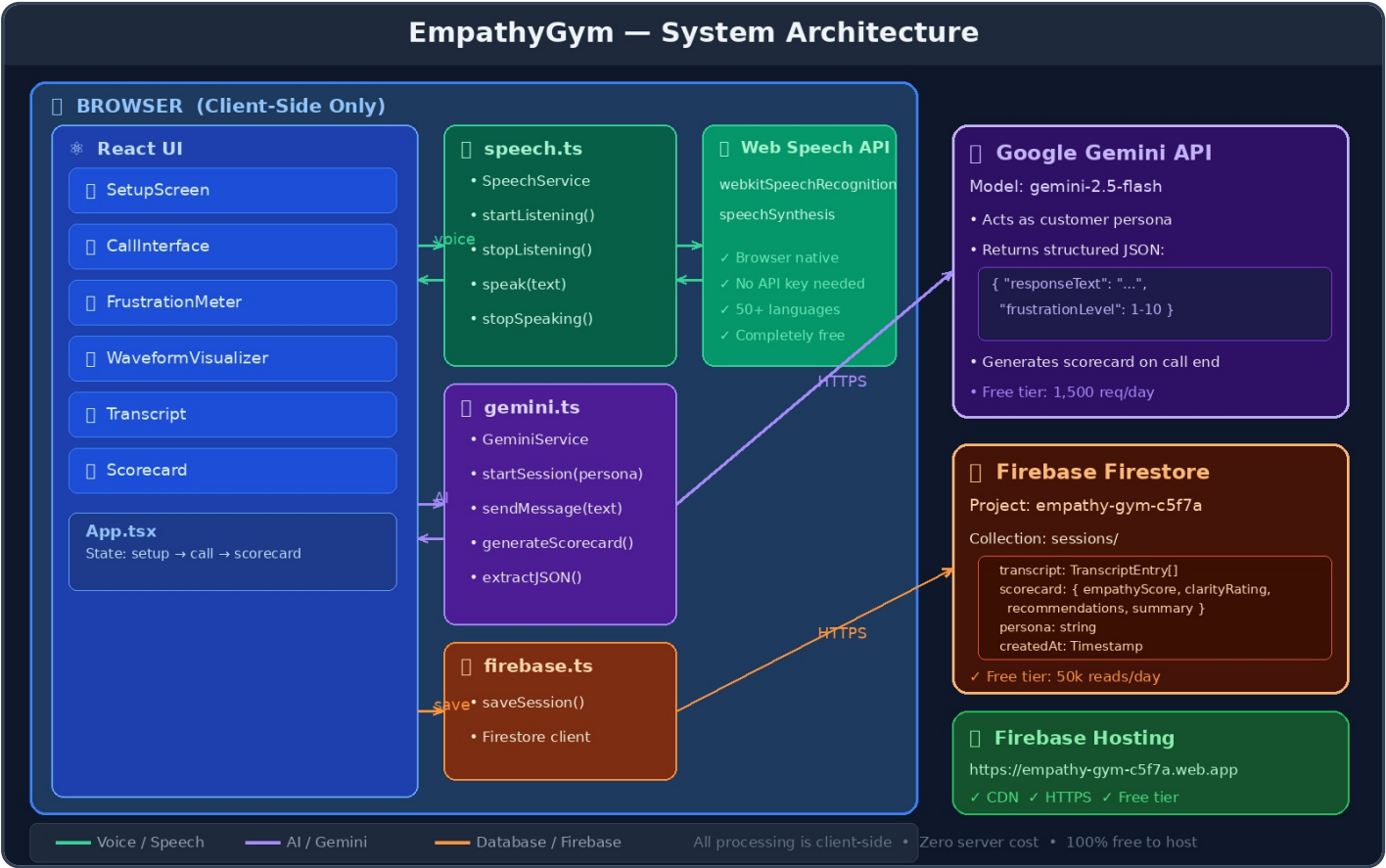
Experienced agents use EmpathyGym to sharpen skills, tackle harder personas, and compete on the leaderboard — turning skill development into a daily habit rather than a one-time event.

Remote & Distributed Teams

Teams spread across time zones train asynchronously. No trainer needs to be online. Sessions are logged to Firestore so managers can review scores and full transcripts at any time from anywhere.

System Architecture

EmpathyGym is a fully client-side application. All AI reasoning, voice processing, and state management happen in the browser. The only external calls are to Google Gemini API (AI) and Firebase Firestore (storage).



Data Flow — Single Turn

Every time the agent speaks, this sequence executes end-to-end in under 2 seconds.

- 1

Agent presses Speak
Browser activates `webkitSpeechRecognition`. Interim results stream into the transcript in real time.
- 2

Final transcript captured
On silence detection, the final transcript is locked and passed to `GeminiService.sendMessage()`.
- 3

Gemini processes the message
The full conversation history + system persona prompt is sent to Gemini 2.5 Flash. It responds as the customer character and returns structured JSON: `{ responseText, frustrationLevel }`.
- 4

UI updates instantly
`FrustrationMeter` animates to the new level. Customer response appends to the transcript.
- 5

Browser speaks the response
`speechSynthesis.speak()` reads the customer's reply aloud. The agent listens and prepares their next response.
- 6

End condition check
If `frustrationLevel ≤ 1` → success (customer calmed). If `≥ 10` → failure (customer hangs up). Either triggers scorecard generation.

Performance Scorecard

At session end, the full transcript is evaluated by Gemini and three metrics are returned.

0–100%

Empathy Score

Did the agent acknowledge feelings, apologise sincerely, and validate the customer's frustration?

0–100%

Clarity Rating

Were responses clear, jargon-free, and solution-focused without overwhelming the customer?

x3

Recommendations

Three specific, actionable coaching tips derived directly from the actual conversation transcript.

Tech Stack

Every layer is free. No credit card, no server, no cold starts.



React 18 + TypeScript + Vite

Fast HMR dev experience. Type-safe components. Production build outputs to dist/.

Web Speech API

Browser-native speech recognition & synthesis. No API key. Works in Chrome. 50+ languages.

Google Gemini 2.5 Flash

Powers customer personas and scorecard generation. Free tier: 1,500 requests/day.

Firebase Firestore

Stores session transcripts and scores. Free Spark plan: 50k reads + 20k writes/day.

Firebase Hosting

Global CDN, HTTPS, custom domains. Free tier: 10 GB storage, 360 MB/day transfer.

Total Monthly Cost

\$0.00

Key Benefits

Benefit	Detail	Impact
Zero Cost	Browser APIs + free tiers only	No budget approval needed
Instant Feedback	Frustration meter updates after every response	Faster learning loop
Safe to Fail	No real customers affected	Agents experiment freely
Infinitely Scalable	No infrastructure to provision	1 or 10,000 agents
Objective Scoring	AI-generated, bias-free evaluation	Fair performance tracking
Async & Self-Paced	No scheduling, no trainer required	Train anytime, anywhere

Future Enhancements

Planned features to extend EmpathyGym into a full-scale training platform.

- Custom Personas**
Managers define their own customer scenarios matching real complaint patterns from their business, uploaded directly in the app.
- Team Leaderboard**
Aggregate empathy scores across the team pulled from Firestore, driving healthy competition and recognising top performers.
- Session Replay**
Play back the full audio and transcript of any saved session for self-review or manager-led coaching conversations.
- Manager Dashboard**
Analytics view showing score trends, weakest personas, and improvement over time per agent — exportable as CSV.
- Multilingual Support**
Web Speech API supports 50+ languages. Extend personas and scoring prompts to support non-English support teams globally.
- Difficulty Progression**
Adaptive difficulty that increases persona aggression as agents improve, ensuring continuous challenge and skill growth.